Davi Guanabara de Aragão

Profile

Work focused on intelligent systems with emphasis on decision problems and machine learning. Experience in reinforcement learning, simulation-driven engineering, and applied research for autonomous systems.

Contact

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Personal Information

Citizenship: **Brazilian**
Address: Sao Paulo, Brazil
Languages: **English (CEFR B2, DET score: 115)**

Skills

- Python (PyTorch)
- Reinforcement and supervised learning in simulation-driven settings
- Deep learning
- Modeling and experimentation in simulated environments

Projects

SkyWeaver

◇ Dynamic airspace configuration platform for UTM/U-Space. ◇ Primary research platform for dynamic airspace configuration. ◇ The public portfolio currently exposes the verified project scope and keeps the structure ready for future public architecture material.

Reed

◇ Reactive runtime for deterministic dataflow execution. ◇ A runtime-oriented project centered on deterministic reactive computation. ◇ The structure is ready for deeper technical documentation as public repository materials become available.

OndeVale

◇ Web application that estimates the price per square meter of real estate in Sao Paulo using only geographic location. ◇ Developed as a data science case study for PO-235 at ITA and UNIFESP, with emphasis on methodology, validation, and deployment. ◇ The public repository separates dataset handling, KNN training artifacts, a FastAPI backend, and a Vue frontend for interactive map-based inference.

Deep Reinforcement Learning with Graph Attention for Cooperative Loyal Wingmen

◇ Published research on cooperative loyal wingmen under shared spatio-temporal observations. ◇ Research focused on multi-agent autonomy with graph attention mechanisms and richer situational awareness. ◇ The publication combines supervised pre-training with reinforcement-learning fine-tuning and uses graph attention networks to process shared observations.

Deep Learning for Geo-Temporal Environmental Modeling

◇ Spatio-temporal data analysis and simulation framework using deep attention mechanisms and uncertainty-aware supervised learning workflows. ◇ Framework for simulating, analyzing, and predicting spatiotemporal phenomena from sparse sensor data. ◇ The framework centers on SensorManager, a Gymnasium-style environment, a wrapper-inspired training interface, and deep learning models built around sets of random neighbors and attention.

Education

Ph.D. in Electronic Engineering and Computing (EEC) - Informatics

Aeronautics Institute of Technology (ITA) **2025 - Present** ◇ Current doctoral research in data science, intelligent systems, and engineering.

Graduate Certificate in Data Science (CEDS)

Aeronautics Institute of Technology (ITA) **2024 - 2026** ◇ Graduate coursework in data science.

Master's in Aeronautical Computing (MPCOMP)

Aeronautics Institute of Technology (ITA) **2021 - 2024** ◇ Research-oriented work in deep reinforcement learning.

Graduate Certificate in Business Management

Fundacao Getulio Vargas (FGV) **2019 - 2021** ◇ One of Brazil's leading institutions in business, economics, and public administration.

Bachelor's Degree in Control and Automation Engineering

University of Fortaleza (UNIFOR) **2013 - 2019** ◇ Undergraduate foundation in control, automation, and embedded systems.

Publications

Deep Reinforcement Learning with Graph Attention for Coopera-